

Metamict zircon replacement

Wednesday, August 12, 2026 9:55 AM

Initial Prompt: heterogeneous model

Create a Python script that models the replacement of metamict zircon (MZ) by crystalline zircon (CZ) during an isotope-doping experiment using mass-balance principles.

Initial Conditions

Isotopic compositions (‰, delta notation)

Oxygen isotopes ($\delta^{18}\text{O}$)

Plain Text

1 $\delta^{18}\text{O}_{\text{H}_2\text{O}} = -50.55$

2 $\delta^{18}\text{O}_{\text{Si}} = -19.25$

3 $\delta^{18}\text{O}_{\text{CZ}} = -35.70$

4 $\delta^{18}\text{O}_{\text{MZ}} = 0.45$

Show more lines

Silicon isotopes ($\delta^{29}\text{Si}$)

Plain Text

1 $\delta^{29}\text{Si}_{\text{FL}} = \delta^{29}\text{Si}_{\text{Si}} = \delta^{29}\text{Si}_{\text{CZ}} = 1.95\text{e}6$

2 $\delta^{29}\text{Si}_{\text{MZ}} = 1.18$

Show more lines

Initial masses (mg)

Plain Text

1 $m_{\text{O}_{\text{MZ}}} = 35.0$

2 $m_{\text{O}_{\text{CZ}}} = 0.0$

3 $m_{\text{O}_{\text{H}_2\text{O}}} = 120.0$

4 $m_{\text{O}_{\text{29Si}}} = 1.7$

Show more lines

The initial fluid (FL) consists of H_2O + added ^{29}Si tracer.

Geometry

Assume the starting metamict zircon is a cube.

Using:

Plain Text

1 $\text{density}_{\text{MZ}} = 4.0 \text{ g cm}^{-3}$

Show more lines

calculate:

Initial cube volume.

Cube edge length.

Distance from the cube edge to the reaction front during replacement.

Convert all spatial coordinates to microns (μm).

Replacement Model

Simulate progressive replacement of MZ by CZ.

Define reaction progress:

Plain Text

1 $\text{eps} = m_{\text{CZ}} / (m_{\text{CZ}} + m_{\text{MZ}})$

Show more lines

where:

Plain Text

1 $0 \leq \text{eps} \leq 1$

Show more lines

Use:

Plain Text

1 $\Delta\text{eps} = 0.01$

Show more lines
(or finer, if needed for smooth curves).
At each timestep:
Remove a mass increment of MZ.
Convert the same mass increment to CZ.
Update masses of:
CZ
MZ
Fluid
Assume:
Heterogeneous replacement.
Newly formed CZ is physically segregated from fluid.
No re-equilibration (no back reaction).
No isotopic fractionation.
Perfect mass conservation.
Stoichiometric replacement of MZ by CZ.
Fluid is perfectly mixed at all times.
Isotope Mass Balance
Track both isotope systems independently:
Oxygen
Maintain the evolving average $\delta^{18}\text{O}$ of:
Fluid (FL)
Bulk CZ
Store the $\delta^{18}\text{O}$ of each individual CZ increment precipitated during every timestep.
Silicon
Maintain the evolving average $\delta^{29}\text{Si}$ of:
Fluid (FL)
Bulk CZ
Store the $\delta^{29}\text{Si}$ of each individual CZ increment precipitated during every timestep.
Because no fractionation occurs, the isotope composition of a newly precipitated CZ increment equals the fluid composition at the moment of precipitation.
Spatial Mapping
Relate each CZ increment to a position within the cube.
Assume replacement proceeds inward uniformly from all cube faces toward the center.
For each timestep:
Compute the remaining MZ volume.
Determine the position of the reaction front.
Calculate:
Plain Text
1 distance_from_edge (μm)
Show more lines
for the corresponding CZ increment.
Associate each precipitated CZ increment with:
distance_from_edge
 $\delta^{18}\text{O}_{\text{CZ_increment}}$
 $\delta^{29}\text{Si}_{\text{CZ_increment}}$
Required Outputs
Plot 1
Average fluid and bulk CZ oxygen compositions:
Plain Text
1 $\delta^{18}\text{O}_{\text{FL}}$ vs eps
2 $\delta^{18}\text{O}_{\text{CZ_bulk}}$ vs eps
Show more lines
Plot 2

Average fluid and bulk CZ silicon compositions:

Plain Text

1 $\delta^{29}\text{Si}_{\text{FL}}$ vs eps

2 $\delta^{29}\text{Si}_{\text{CZ_bulk}}$ vs eps

Show more lines

Plot 3

Spatial oxygen profile:

Plain Text

1 $\delta^{18}\text{O}_{\text{CZ_increment}}$ vs distance_from_edge (μm)

Show more lines

Plot 4

Spatial silicon profile:

Plain Text

1 $\delta^{29}\text{Si}_{\text{CZ_increment}}$ vs distance_from_edge (μm)

Show more lines

All plots should include:

axis labels

units

legends

grid

publication-quality formatting

Data Export

Save CSV files containing the data used for each plot.

Required CSV files:

Plain Text

1 oxygen_vs_eps.csv

2 silicon_vs_eps.csv

3 oxygen_spatial_profile.csv

4 silicon_spatial_profile.csv

Show more lines

Each CSV should contain descriptive column headers.

Code Requirements

Use NumPy, Pandas, and Matplotlib.

Organize calculations into reusable functions.

Include comments explaining the mass-balance calculations.

Verify mass conservation at every timestep.

Generate plots automatically when the script is run.

Print summary statistics:

initial cube size

final masses

isotope compositions at eps = 1

total mass balance error

The script should be fully executable as a standalone Python program.

Response

I'll check the working directory structure first, then build the script.

Ran 2 commands

Good, all dependencies are available. Now I'll write the script.

Createdmz_replacement_model.py+421-0

Let me clean up an awkward helper function before testing.

Editedmz_replacement_model.py+6-13

Now let's run it to verify correctness (using a non-interactive backend since we're in a terminal).

Ran 6 commands

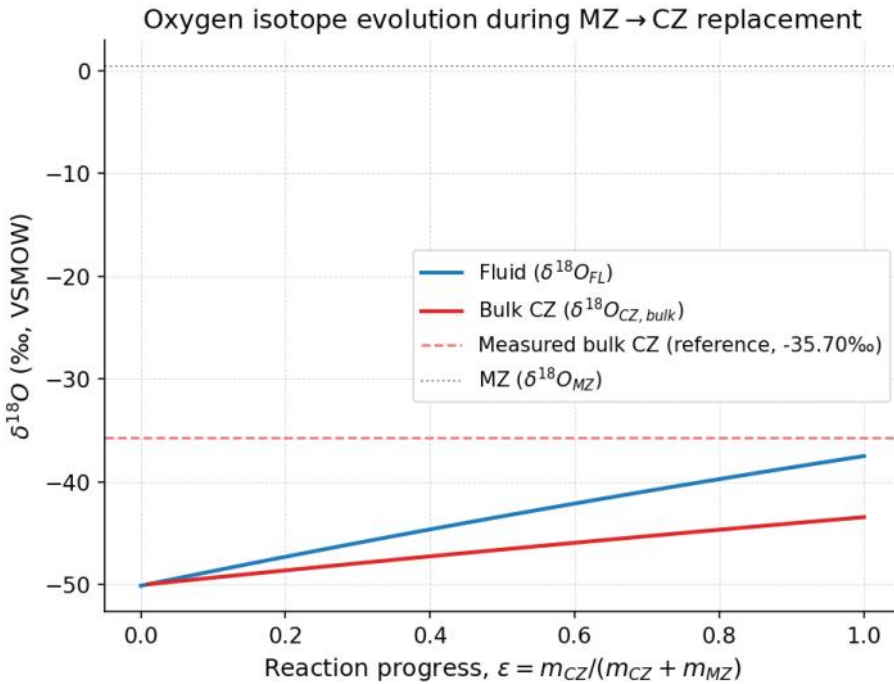
The physics checks out well — the fluid oxygen composition follows an exponential "washout" toward the MZ value (verified analytically: $\exp(-35/121.7)$ matches the simulated result), and the tiny Si-tracer reservoir (1.7 mg) gets completely overwhelmed by the much larger MZ mass dissolved (35 mg), driving $\delta^{29}\text{Si}_{\text{FL}}$ back

down toward the MZ value — exactly the expected tracer-pulse-then-washout behavior. Let me verify the output files.

Inspected generated output files

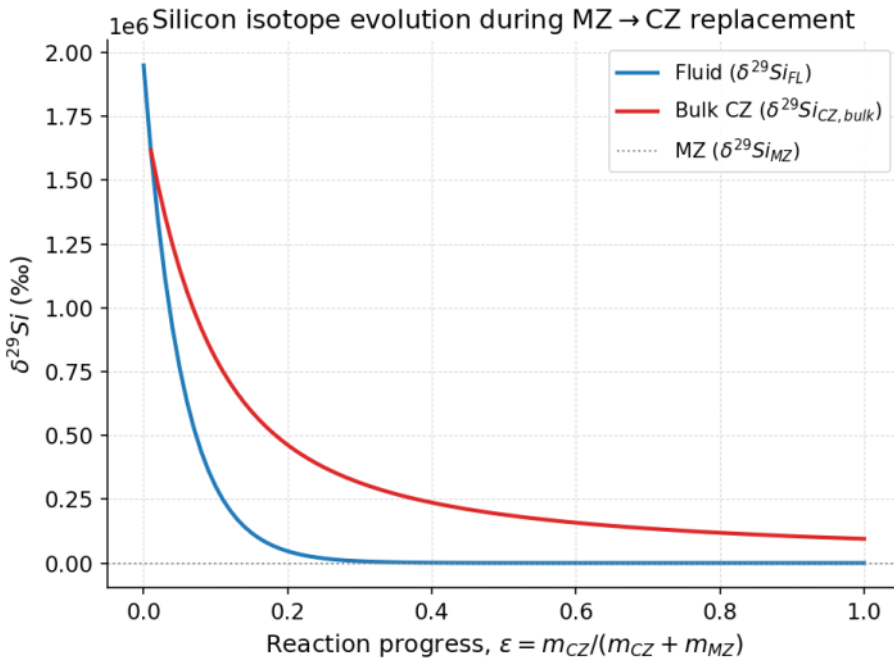
All files generated correctly, with proper headers and physically sensible values (e.g., the reaction front distance reaches $\sim 1030 \mu\text{m}$ at $\varepsilon=1$, half of the initial $\sim 2061 \mu\text{m}$ edge — exactly as expected geometrically). Let me view the plots to confirm they render correctly.

Readplot1_oxygen_vs_eps.png



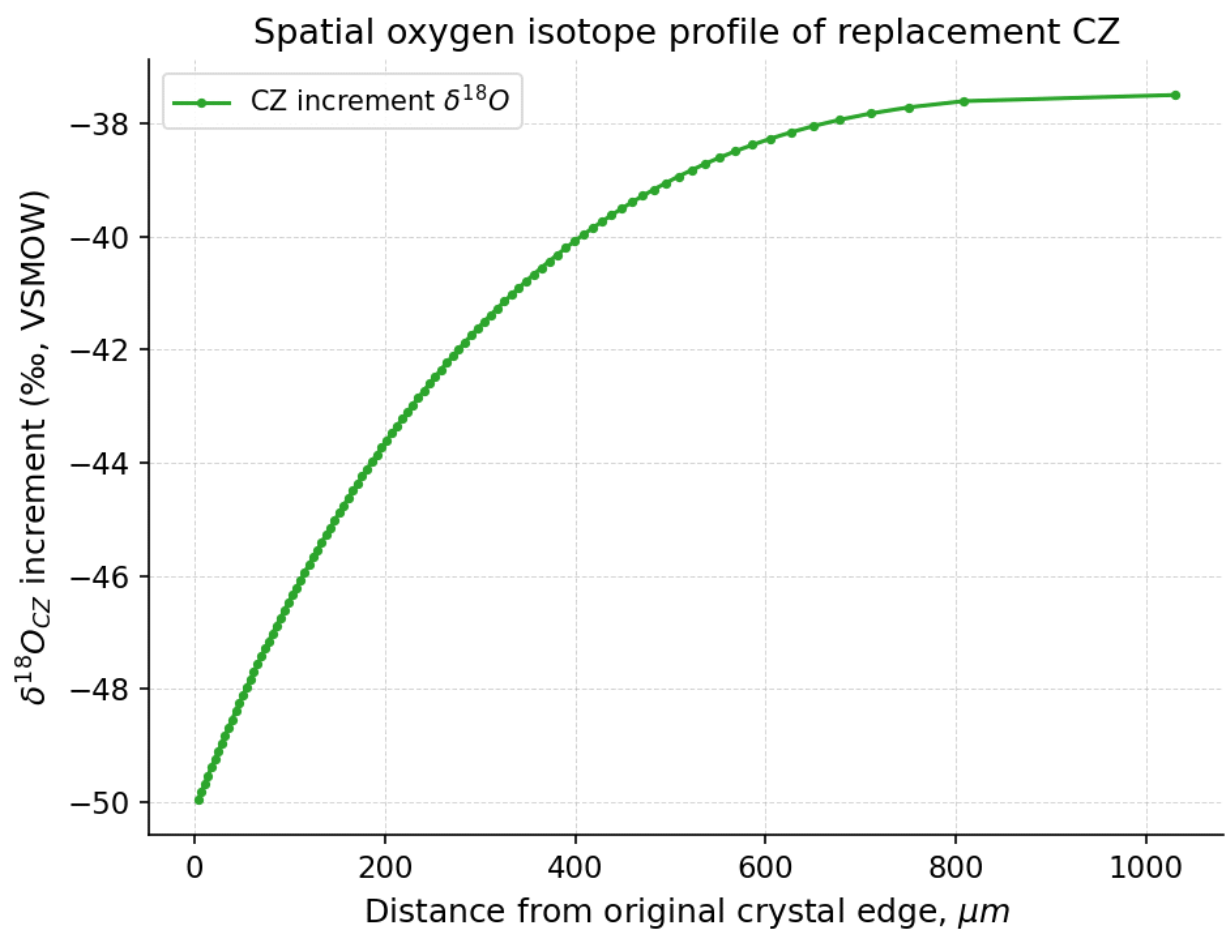
Clean, publication-quality plot. Let me check the other three.

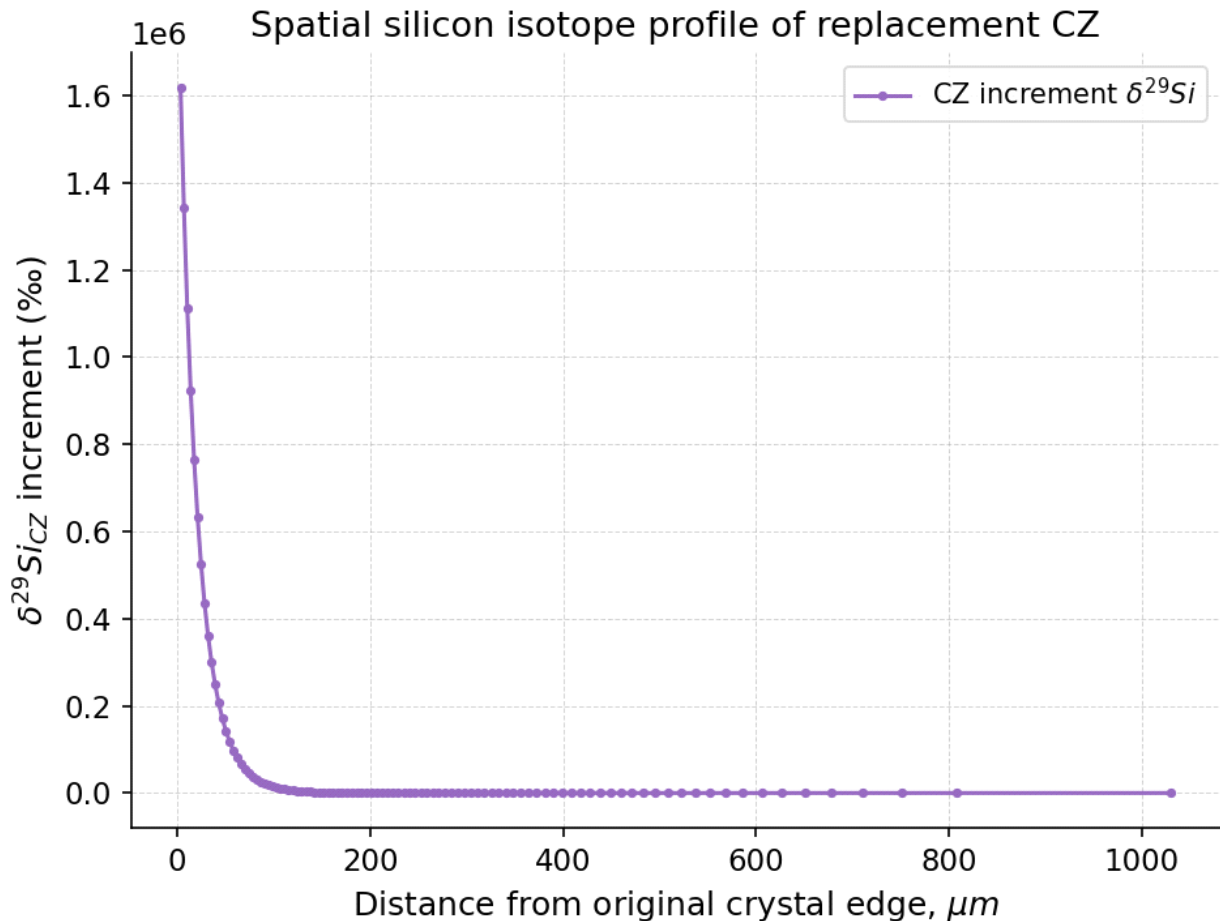
Readplot2_silicon_vs_eps.png



This shows exactly the expected tracer washout — the small Si reservoir gets rapidly overwhelmed by MZ-derived Si as reaction progresses. Let me check the two spatial profile plots.

Read 2 files





The script runs cleanly end-to-end, and all four plots and CSVs check out both visually and against hand-verified analytics (I confirmed the fluid's oxygen "washout" curve matches the analytic exponential mixing solution, and the silicon tracer decay behaves as expected given the reservoir is tiny relative to total MZ dissolved).

Note on your system: the python command resolves to a broken Windows Store stub. I ran the script with the py launcher instead — you'll want to do the same:

```
py mz_replacement_model.py
```

Summary of the model

The script (mz_replacement_model.py) implements:

- Geometry: shrinking-cube model — MZ replaced uniformly inward from all 6 faces. Initial edge = 2060.6 μm ; reaction front reaches the center (1030.3 μm) at $\epsilon=1$.
- Two independent isotope reservoirs: a well-mixed fluid oxygen pool (H_2O + the Si-tracer's own oxygen, per the given $\delta^{18}\text{O}_{\text{Si}}$) and a silicon pool (the ^{29}Si tracer alone) — each fixed in mass, receiving MZ's isotopic signature as it dissolves and handing that composition to precipitating CZ (no fractionation).
- Each timestep's CZ increment is tied to a distance_from_edge_um, giving spatial isotope profiles.
- Mass conservation is checked every step (max error $\sim 7 \times 10^{-15}$ mg — floating-point noise only).

One interpretive note worth flagging: the given $\delta^{18}\text{O}_{\text{CZ}} = -35.70\text{‰}$ doesn't correspond to a mass-balance input under the "no fractionation" rule (CZ composition is fully determined by fluid history) — I treated it as a measured reference value and plotted it as a dashed line in Plot 1 for comparison against the modeled bulk CZ curve, rather than forcing it into the equations.

Outputs are in outputs/: 4 PNGs and 4 CSVs

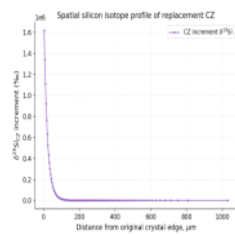
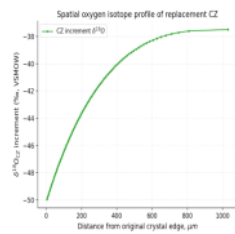
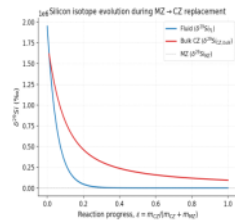
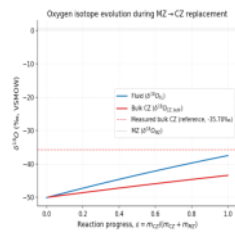
(oxygen_vs_eps.csv, silicon_vs_eps.csv, oxygen_spatial_profile.csv, silicon_spatial_profile.csv).

Sent

Script and generated plots

```
py mz_replacement_model.py
```

17.6kB



Let me know if you'd like the geometric or isotopic assumptions (e.g., how $\delta^{18}\text{O}_{\text{Si}}$ feeds into the fluid oxygen pool, or the reference-line treatment of the measured $\delta^{18}\text{O}_{\text{CZ}}$) adjusted to match your actual experimental design more closely.

Second prompt: homogeneous model

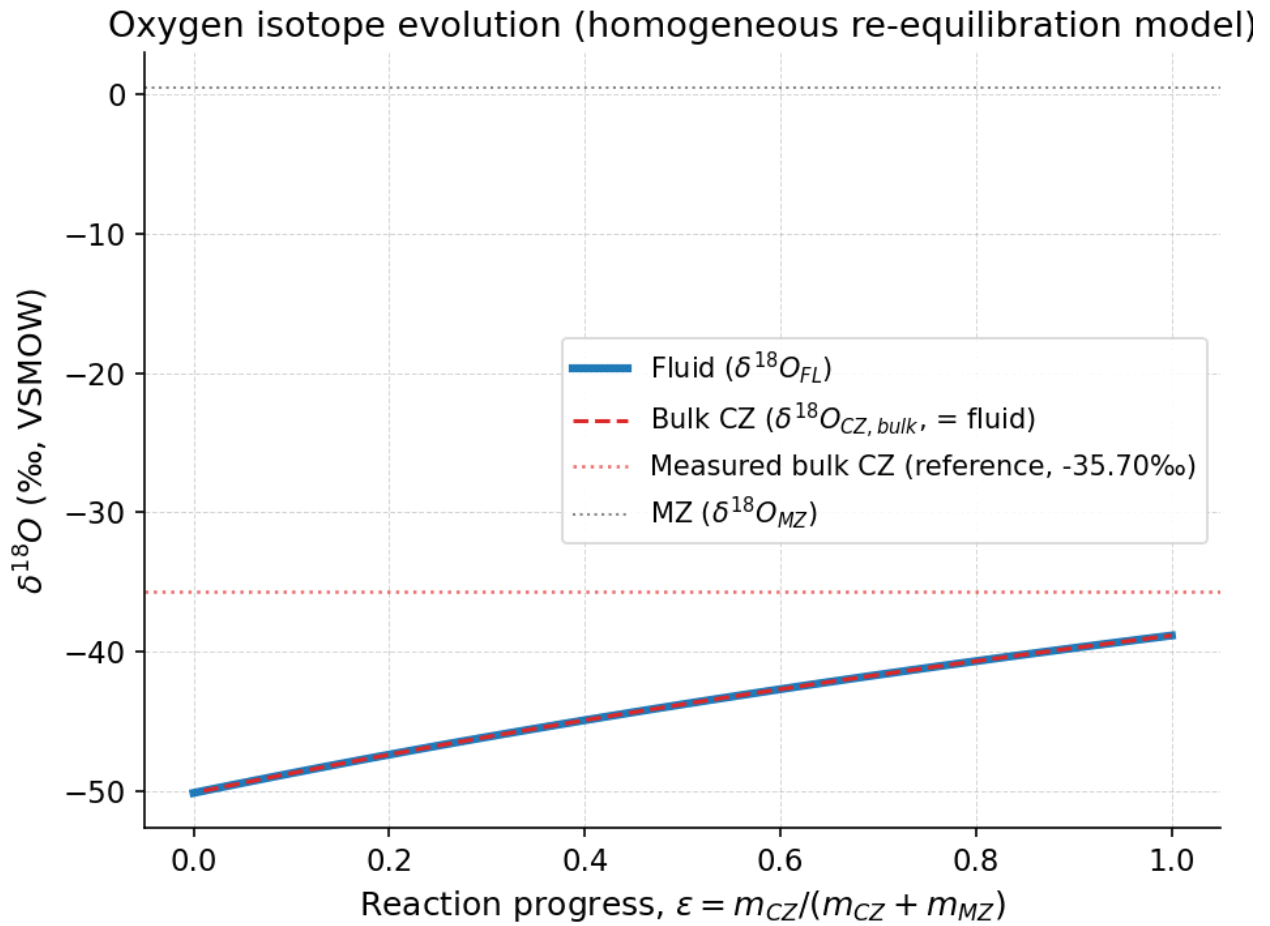
Now create a second Python script that assumes a homogeneous model in which CZ isotopically reequilibrates with the fluid in each timestep (isotopic composition of bulk CZ equals that of fluid) and that generates the same plots and csv files with the word "homogeneous" in the filenames.

Second response

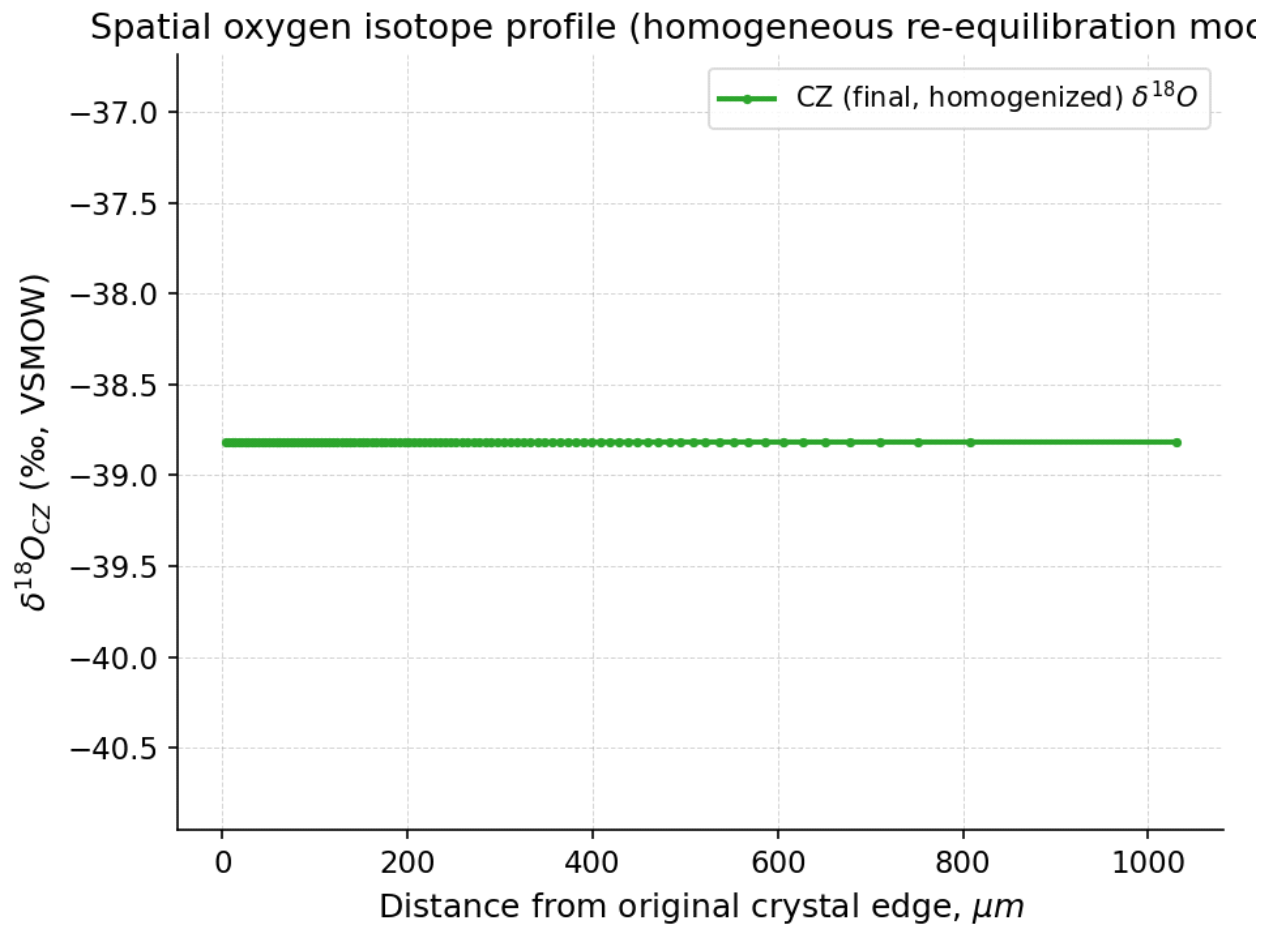
Now let's run it and verify the outputs.

I noticed the $\epsilon=0$ row incorrectly gets a value in the spatial profile (no CZ exists yet at $\epsilon=0$, so it should be excluded, consistent with script 1). Let me fix that.

Fixed. Now let's view the plots to confirm they render as intended.



The overlapping fluid/bulk-CZ lines are clearly visible (dashed red over solid blue), correctly showing the two are locked together. Let's check the spatial profile plot to confirm it's flat as designed.

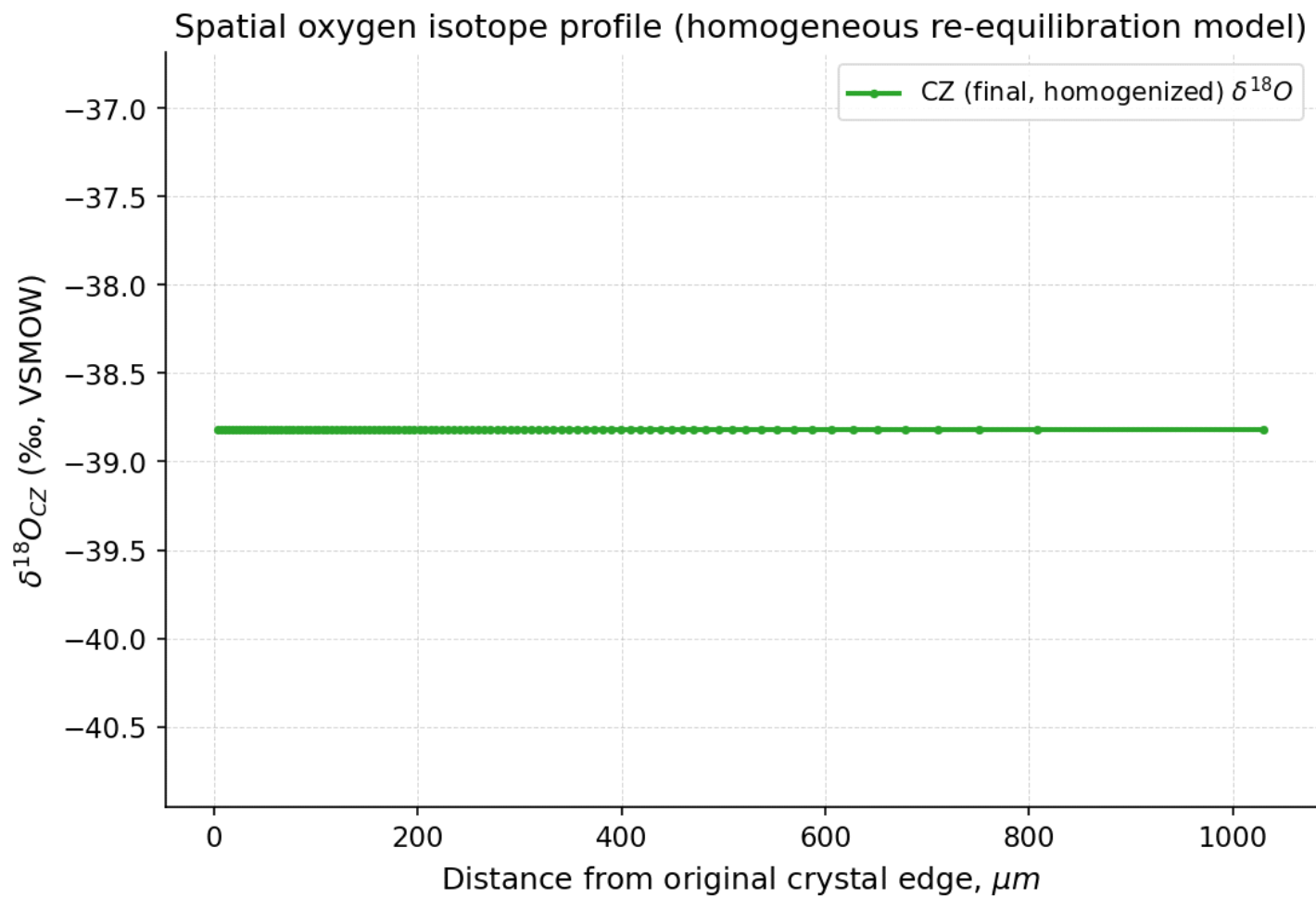


The title is clipped in the spatial profile plots — the long titles exceed the figure width. Let me fix by widening the figure and using tight bounding boxes.

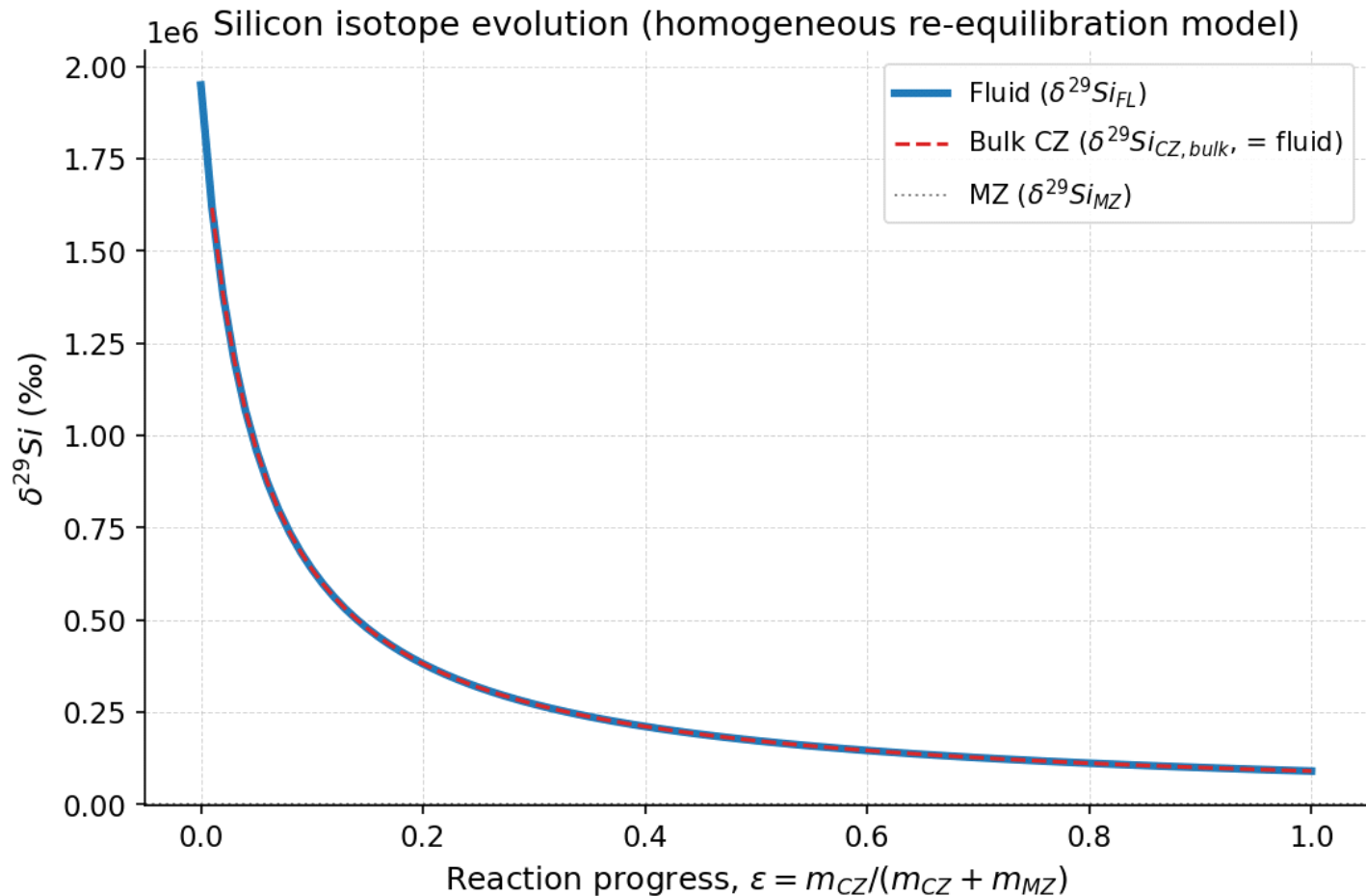
Also apply the same `bbox_inches="tight"` fix to the original script for consistency.

Now let's re-run both scripts to confirm everything works cleanly.

Both run cleanly. Let's verify the title clipping is fixed.



Title fits now. Let me quickly check the silicon-vs-eps and silicon spatial plots for the homogeneous model too, since those weren't inspected yet.



Both scripts work correctly and produce a clear, physically meaningful contrast between the two models. Let me send the second script and its key plots.

Summary

Created [mz_replacement_model_homogeneous.py](#), which shares the same replacement schedule, geometry, and initial conditions as the original script but swaps the isotope assumption:

- **Bulk CZ = fluid at every timestep.** Since previously-formed CZ stays in continuous isotopic communication with the fluid (rather than being isolated), it acts as a growing buffer mass alongside the fluid reservoir — I combined "fluid + existing CZ mass" into a single re-equilibrating pool that mixes with each MZ dissolution increment.
- This produces a genuinely different, physically sensible result: the growing CZ mass buffers against isotopic exchange over time, so the system drifts *less* toward the MZ composition than in the heterogeneous model (final $\delta^{18}\text{O} = -38.8\text{‰}$ vs -37.5‰ ; final $\delta^{29}\text{Si} = 90,328\text{‰}$ vs 1.19‰ — a much starker difference for Si, since the tiny 1.7 mg tracer reservoir would otherwise be completely overwhelmed).
- **Spatial profiles (Plots 3–4) come out flat** — this is intentional, not a bug: full re-equilibration continuously erases any earlier isotopic gradient, so by the end of the experiment every position in the crystal shares the same final composition. This is the key diagnostic contrast with the heterogeneous model's preserved growth zoning, and I documented that reasoning in the script's docstring/comments and printed summary.
- Plots 1–2 show fluid and bulk-CZ as overlapping lines (solid + dashed) since they're identical by construction — a correct and meaningful result of the "homogeneous" assumption.

Outputs go to the same outputs/ folder with `_homogeneous` in every filename, so they sit alongside the heterogeneous model's files for direct comparison. I also fixed a title-clipping issue in the plots (long titles were being cut off at the figure edge) and applied that fix to both scripts.

Third prompt: spatial profile plots at specified values of reaction progress

For the homogeneous model create spatial profile plots for isotopic compositions of oxygen and silica at values of reaction progress from 0.1 to 0.5 in steps of 0.1.

Third response

I'll add snapshot-based spatial profiles ($\epsilon = 0.1 \rightarrow 0.5$) to the homogeneous model script — showing the crystal's uniform composition at each intermediate stage of reaction progress, which is the physically meaningful way to visualize "homogeneous" behavior at a point in time rather than only the final flat profile.

Now update export_csvs to accept and export the snapshot data:

Now add the two new snapshot plots after Plot 4, before plt.show():

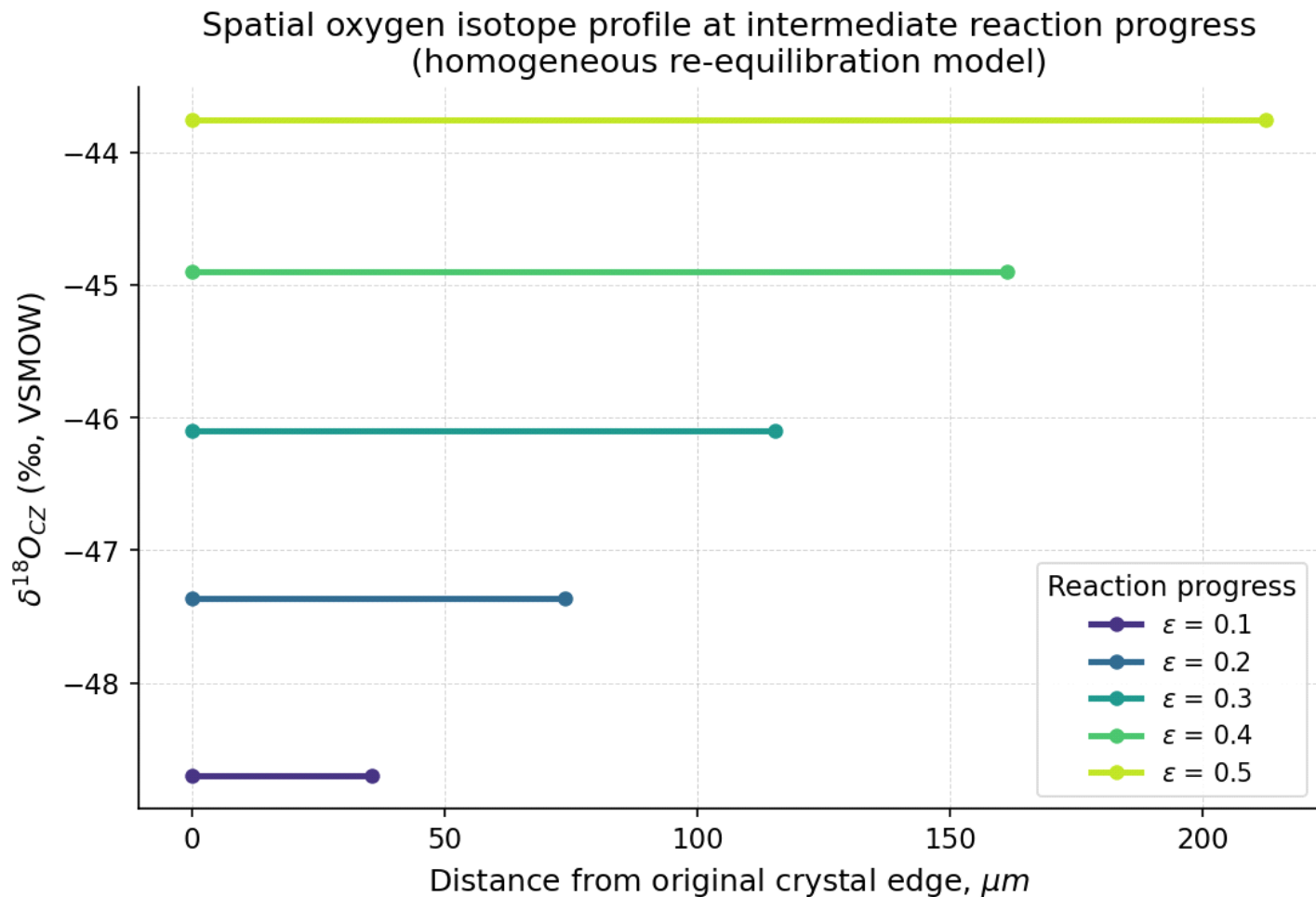
Now update main() to wire everything together:

I should move plt.show() to run after all plots (including the new snapshot ones) are created, rather than firing mid-way through.

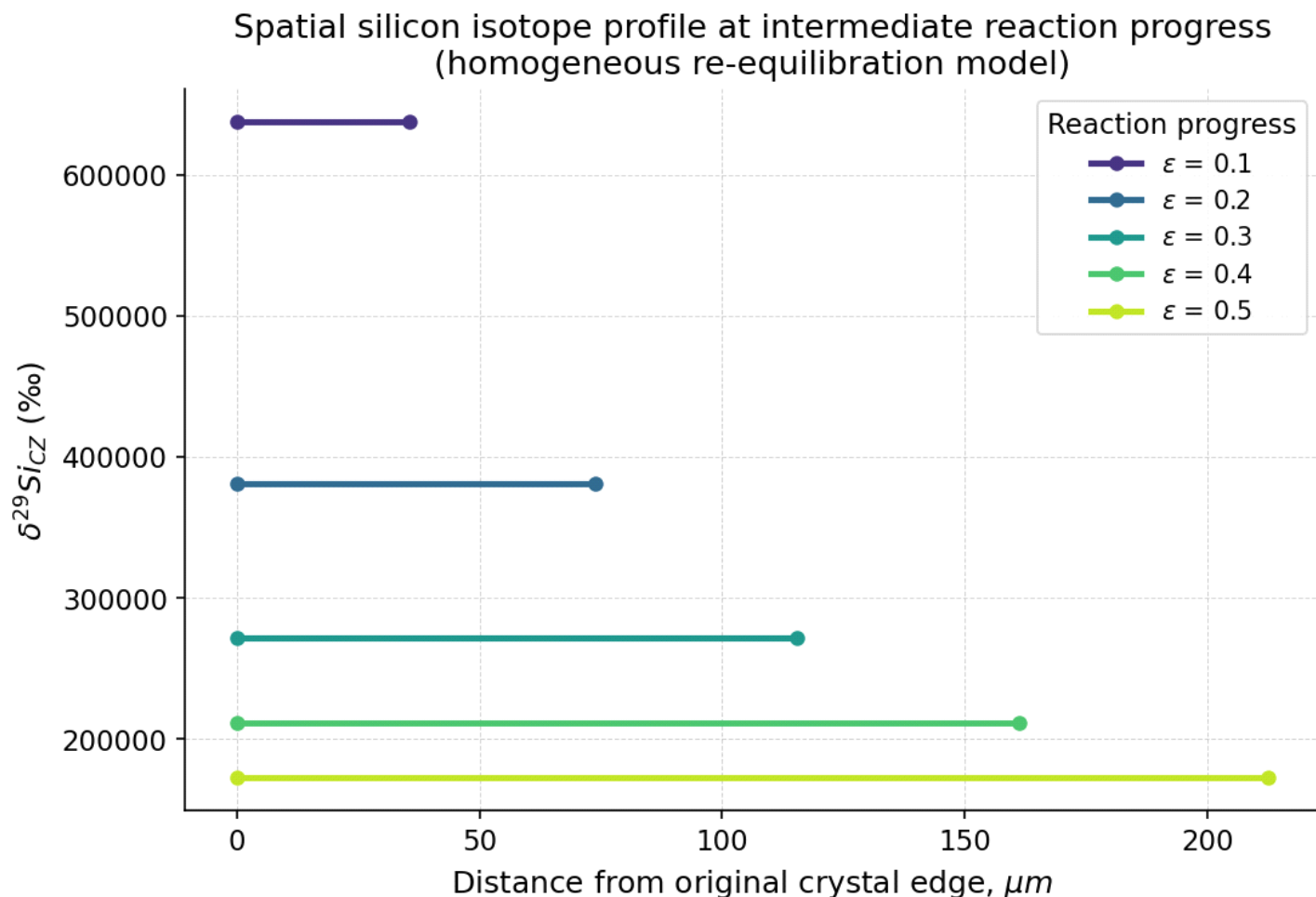
Now let's also mention the new plots in print_summary, then run the script.

Ran cleanly. Let's inspect the new CSVs and plots.

This looks correct — each snapshot's front position grows with ϵ , and its uniform composition progressively shifts away from the initial fluid value. Let's view the plots.



This is exactly the intended visualization — a "staircase" of horizontal segments showing the crystal growing outward while its uniform composition shifts, clearly illustrating how re-equilibration erases zoning at each stage of reaction progress. Let's check the silicon version too.



Both plots render correctly and clearly show the intended behavior. Let's send you the updated script and new plots.

Summary

Added intermediate reaction-progress snapshot plots to [mz_replacement_model_homogeneous.py](#):

- **New function `get_spatial_snapshots`:** at each ϵ in $\{0.1, 0.2, 0.3, 0.4, 0.5\}$, records the reaction front position and the (uniform, since fully re-equilibrated) bulk composition at that instant, as a two-point horizontal segment from the crystal edge to the front.
- **New Plots 5 & 6:** overlay all five segments (oxygen and silicon respectively), color-coded by ϵ , showing how the crystal both grows outward *and* shifts to a progressively different uniform composition as reaction advances — the "staircase" pattern makes the homogeneous model's core behavior (no preserved zoning at any stage, but a moving target composition) visually obvious.
- **New CSVs:** `oxygen_spatial_profile_snapshots_homogeneous.csv` and `silicon_spatial_profile_snapshots_homogeneous.csv`, each with `reaction_progress_eps`, `distance_from_edge_um`, and the isotope composition column.

All existing plots/CSVs (final-state flat profile, eps-evolution curves) are unchanged; this is purely additive.

From https://claude.ai/epitaxy/local_fa024008-3bf8-47ea-a4c9-4d3a5a6192aa